

CS 171 / EE 142: Intro to ML and DM

Christian Shelton

UC Riverside

Slide Set 0: Syllabus



- From UC Riverside
 - ▶ CS 171 / EE 142: Introduction to Machine Learning and Data Mining
 - ▶ Professor Christian Shelton
- DO NOT REDISTRIBUTE
 - ▶ For use only by enrolled students in the course

Lecture:

Mon, Wed, Fri
3:00pm–3:50pm
Spieth 2200

Discussion:

Tue
1:00–1:50pm (ND A1010)

Instructor:

Christian Shelton
cshelton@cs.ucr.edu
off hrs: Thursdays, 4pm–6pm
MRB 4118

TA:

Xiaoyin Pan
xiaoyin.pan@email.ucr.edu

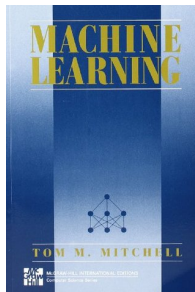
Prerequisites:

- Linear Algebra (Math 31, EE 20B)
- Multivariate Calculus (Math 10A)
- Intro Prob/Stats (Stat 155, EE 114)
- Solid Programming (CS 100, EE 120B)

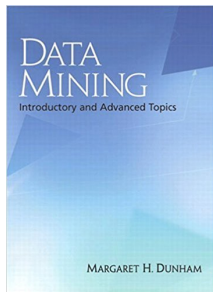
Preprints of a new book will be provided

Possible (all optional) Real Texts:

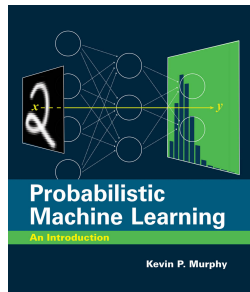
Machine Learning
by Tom M. Mitchell



Data Mining: Introductory and Advanced Topics
by Margaret H. Dunham



Probabilistic Machine Learning: An Introduction
by Kevin P. Murphy



Syllabus

Monday	Wednesday	Friday	Due
3/30 Introduction [1,2]	4/1 Bayes-optimal [3]	4/3 Naive Bayes [3]	
4/6 Linear regression [4]	4/8 Linear regression [4]	4/10 Perceptron [5]	PS 1
4/13 Logistic Regression [5]	4/15 Logistic Regression [5]	4/17 Features [6]	Q1 [3–5]
4/20 Regularization [6]	4/22 Validation [6]	4/24 SVM [7]	PS 2
4/27 SVM [7]	4/29 Kernels [7]	5/1 Nearest Neighbor [9]	Q2 [6–7]
5/4 Normalization [9]	5/6 Neural Networks [8]	5/8 Neural Networks [8]	PS 3
5/11 Neural Networks [8]	5/13 Decision Trees [10]	5/15 Decision Trees [10]	Q3 [8–9]
5/18 Decision Trees [10]	5/20 Clustering [11]	5/22 Clustering [11]	PS 4
5/25 <i>Memorial Day</i>	5/27 Feature Selection [12]	5/29 PCA [12]	Q4 [10–12]
6/1 PCA [12]	6/3 Data Representations [13]	6/5 Review	
Final Exam: Monday, June 8, 8:00am–11:00am			

Syllabus

- Course Work:
 - ▶ 4 problem sets (20 pts each)
 - ▶ 4 quizzes (20 pts each)
 - ▶ 1 final (40 pts)
- Problem Sets
 - ▶ Due Friday night (11:59pm)
 - ▶ Do parts throughout the prior weeks!
 - ▶ 4 late days to be used on problem sets
 - ▶ programming in Python notebook, at <https://hub.cs.ucr.edu/>,
 - ▶ plus non-programming questions
 - ▶ **no external sources (friends, web, etc), except on clarification on Python**
- Tests
 - ▶ In CS Testing Center
 - ▶ Quizzes: 50 minutes, sometime on Friday
 - ▶ Final: 3 hours, Monday June 8, 8am-11am
 - ▶ Many end-of-chapter problems were previously test questions
- Sections Tuesdays
- On-line discussions via eLearn

What's this Course About

Machine Learning:

- Grew out of AI
- How can a computer improve based on experience?

Data Mining:

- Grew out of Databases
- How can we make useful information out of a large database?

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Statistics:

- What can we say mathematically about data?

Optimization:

- How can we improve a function, utility, or score